

Homework 4

Due: November 21, 2025

Total Points: 100

Instructions:

- Show all reasoning or calculations where relevant.
- When asked for an interpretation, explain your answer clearly in words.
- You may use R for any applied components. Include all necessary R code and short interpretations.

Q1. (20 points) You are studying a dataset of 500 patients, each with 200 biomarkers measured at baseline. You plan to visualize the relationships between individuals using a low-dimensional embedding (2D or 3D). Consider the following situations and state whether a **global** or **local** dimension reduction method would be more appropriate. Justify your answer in 2–3 sentences.

- (a) You believe there is a single underlying continuous “disease severity” score that explains most of the biomarker variation.
- (b) You are interested in identifying small subgroups of patients with very similar biomarker profiles but distinct from the rest of the population.
- (c) You plan to create a 2D visualization of single-cell RNA expression data to identify cell-type clusters.
- (d) You have time-series physiological data (e.g., heart rate, sleep, and activity) and want to summarize the overall level and variability across participants.

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Q2. (20 points) Complete the table below by summarizing the key ideas for each dimension reduction method.

Method	Main Objective	When to Use / Key Characteristics
PCA		
Factor Analysis		
ICA		
t-SNE		
UMAP		

Q3. (30 points) In this exercise, you will analyze country-level global health indicators from the `gapminder` dataset. This dataset contains country-level statistics including GDP per capita, life expectancy, and population size from 1952–2007.

Step 1: Load the data.

```
# Install/load packages
install.packages("gapminder")
library(gapminder)
data(gapminder)

# Subset the most recent year
gap <- subset(gapminder, year == 2007)

# Keep quantitative variables only
gap_sub <- gap[, c("lifeExp", "gdpPercap", "pop")]
rownames(gap_sub) <- gap$country
```

Step 2: Dimension Reduction Tasks

- Perform a **Principal Component Analysis (PCA)** on the standardized variables. Report the proportion of variance explained by each component and create a biplot showing countries colored by continent.
- Conduct a **Factor Analysis** with two latent factors. Interpret what each factor appears to represent in terms of economic and health dimensions.
- Perform an **Independent Component Analysis (ICA)** with two components. Compare the ICA components to the PCA loadings and discuss differences in interpretation.
- Apply both **t-SNE** and **UMAP** with two dimensions. For each approach, create a scatterplot for each method with points colored by continent. UMAP is supposed to represent more global vs. local structure than t-SNE. Do the results appear to back this statement up? Why?

- (e) Include one paragraph comparing PCA, FA, ICA, t-SNE, and UMAP for this dataset.

Q4. Conceptual and Comparative (15 points)

Complete the table below comparing the following clustering methods.

Method	Clustering Criterion / Model	Key Strengths and Limitations
K-Means		
Gaussian Mixture Model (GMM)		
PAM (Partitioning Around Medoids)		
Hierarchical Clustering		

Q5. Application: Air Quality Site Clustering (15 points)

You will cluster air-quality monitoring sites using publicly available data on multiple pollutants.

Step 1: Load the data.

```
# Install/load packages
install.packages("openair")
library(openair)
data("mydata", package = "openair")
head(mydata)
```

The dataset `mydata` includes daily air-quality observations with columns such as `pm10`, `no2`, `so2`, and `o3`.

Step 2: Create summary features for each monitoring site.

```
library(dplyr)
site_means <- mydata %>%
  group_by(site) %>%
  summarise(across(c(pm10, no2, so2, o3), mean, na.rm = TRUE))
```

Step 3: Clustering Tasks

- (a) Apply **K-Means** clustering to the sites using the four pollutant means. Determine the optimal number of clusters using the `fviz_nbclust()` function from `factoextra`.
- (b) Fit a **Gaussian Mixture Model (GMM)** to the same data and compare the BIC for models with 2–6 clusters.
- (c) Apply **PAM (Partitioning Around Medoids)** and **Hierarchical Clustering** using Euclidean distance. Create and briefly interpret a dendrogram.
- (d) Compare the clustering results across methods. Which sites consistently cluster together?
- (e) How could the clustering results inform air-quality surveillance or intervention planning?